

DTS407TC Large Model Techniques

Group Project – Designing and Evaluating a Domain-Specific Large Model System

Due: Sunday May 17th, 2026 @ 11:59pm

Weight: 40% of the final mark

Group size: 2-3 students

Peer Assessment Weight: 30%

Submission: Report (PDF with maximum 3000 words), Code (zip), Live in-class demonstration (a ppt in pdf format)

Overview

In this coursework, each group is required to design, implement, and evaluate a domain-specific AI system built around a large language model. The objective is to move beyond isolated model training and instead consider how large models are integrated into coherent systems designed to address concrete problems.

Students are expected to define a clearly scoped real-world task, justify their design decisions, and demonstrate how different modelling strategies influence system behaviour. The emphasis of this project lies not only in technical implementation, but also in evaluation, reasoning, and reflection.

This assignment assesses practical competence, analytical thinking, teamwork, and the ability to interpret model performance in realistic settings.

Learning Outcomes

- D. Exhibit proficiency in the practical skills required for implementing and deploying large models, including data preprocessing, model training, finetuning, and result interpretation.
- E. Acquire a thorough knowledge and heightened awareness of the ethical and societal implications linked to large model techniques, addressing biases, fairness, privacy issues, and associated risks.

Team policy

You are free to form your group up to three team members (with a minimum of two members), you need to submit all team members' information before 22nd April, 23:59 via [LMO](#). Students who fail to do so will be randomly formed into teams. Changes will not be allowed once the teams have been confirmed.

A peer assessment, weighted at 30%, will be used to evaluate each individual's contribution. This means 70% of the final grade is based on the group's overall performance, while 30% reflects individual contributions. For more details, please refer to the Peer Assessment section.

Avoid Plagiarism

- Do **not** submit work from other teams.
- Do **not** share code/work to students other than your own team members.
- Do **not** read code/work from other teams, discussions between teams should be limited to high level only.

Project Expectations

Each group must begin by identifying a specific problem situated within a clearly defined domain. The problem should be narrow enough to allow meaningful development and evaluation within the project timeline. Examples might include specialised question answering, domain-adapted summarisation, structured information extraction, controlled generation, or task-oriented assistance systems. Broad ideas such as “a general chatbot” are not acceptable without a well-defined scope.

The system must be built around at least one open-source large language model. Groups may adopt different strategies, including prompt-based methods, retrieval-augmented generation, parameter-efficient fine-tuning, or lightweight domain adaptation. Full-scale training is not required; thoughtful design and justification are more important than model size.

A central requirement of this coursework is comparative reasoning. Each group must implement and analyse at least two modelling strategies within the same problem setting. The comparison may involve, for example, a base model versus a retrieval-enhanced version, prompt engineering versus parameter-efficient fine-tuning, or alternative architectural configurations. The purpose is to understand design trade-offs rather than to optimise a single configuration in isolation.

Evaluation must include both quantitative and qualitative components. Quantitative evaluation may involve task-specific metrics, efficiency measurements, or robustness tests, depending on the nature of the task. Qualitative evaluation should examine system outputs critically, identifying strengths, weaknesses, and failure modes. Groups are expected to interpret their findings rather than simply report numbers.

Deliverables

- Each group is required to submit a written report documenting the full project. The report should describe the motivation (10 marks), system design (10 marks), dataset (10 marks), experiment design (10 marks), evaluation methodology and results (10 marks), and critical analysis on ethical issues such as bias, fairness, and privacy (10 marks). Clear explanation of design decisions is essential. **(60 marks in total)**
- A complete and runnable codebase must also be submitted. The code should be organised and sufficiently documented to allow reproduction of core results. Only key files need to be included; unnecessary intermediate artefacts should be excluded. **(10 marks)**

- During the scheduled presentation session, each group will deliver a concise five-minute presentation introducing the problem, system design, and main findings, followed by a five-minute live demonstration of the working system and a brief question-and-answer session. The demonstration should illustrate typical use cases and, where possible, highlight differences between modelling strategies. **(30 marks in total** for the presentation, live demonstration, and Q&A; see the marking scheme for the breakdown)

Submission

- a **code.zip**: All source code (.py or Jupyter notebook), and
- a **{group-id}-report.pdf**, use the same Latex template provided in A1, and
- a **{group-id}-ppt.pdf**, and
- submit peer assessments on **LMO**.
- Late submission will receive 5% penalties per day, up to 5 working days.
- The assignment must be submitted via Learning Mall to the correct drop box. Only electronic submission is accepted and no hard copy submission. All students must download their file and check that it is viewable after submission. Documents may become corrupted during the uploading process (e.g. due to slow internet connections). However, students themselves are responsible for submitting a functional correct file for assessments.

Peer Assessment

Each group member's contribution will be evaluated through a peer assessment process, conducted anonymously via Core. All group members are required to log in to Core and submit their individual assessments. These marks must be submitted immediately after the group submission is completed. The final grade for each individual will be calculated using the following formula:

$$\text{Final Grade} = (\text{Peer Assessment Weight} \times \text{Student Contribution} \times \text{Group Grade}) + [(1 - \text{Peer Assessment Weight}) \times \text{Group Grade}]$$

Where:

- **Peer Assessment Weight** is 30%.
- **Student Contribution** is determined automatically based on peer scores.

For example, if a student received a peer assessment score of 0 and the group grade is 70, his/her final grade would be calculated as:

$$0.3 \times 0 \times 70 + (1 - 0.3) \times 70 = 49,$$

another student in the same group received peer assessment of 0.95, his/her final grade would be:

$$0.3 \times 0.9 \times 70 + (1 - 0.3) \times 70 = 67.9 \approx 68,$$

Table 1: Peer Review Rubrics

Marks	8-10	5-7	2-4	0-1
Contributions	Routinely provides useful ideas when participating in the group discussion. A leader who contributes a lot of effort.	Usually provides useful ideas when participating in the group discussion. A strong group member who tries hard!	Sometimes provides useful ideas when participating in the group discussion. A satisfactory group member who does what is required.	Rarely provides useful ideas when participating in the group discussion. May refuse to participate.
Problem-solving	Actively looks for and suggests solutions to problems.	Refines solutions suggested by others.	Does not suggest or refine solutions, but is willing to try out solutions suggested by others.	Does not try to solve problems or help others solve problems. Lets others do the work.
Working with others	Almost always listens to, shares with, and supports the efforts of others. Tries to keep people working well together.	Usually listens to, shares with, and supports the efforts of others. Does not cause "waves" in the group.	Often listens to, shares with, and supports the efforts of others, but sometimes is not a good team member.	Rarely listens to, shares with, and supports the efforts of others. Often is not a good team player.

Marking Scheme

Category	Assessment Guidelines	Marks
Motivation	Outstanding (8–10): Clearly defines a well-scoped domain-specific problem, provides strong motivation, and explains why the task is important and suitable for a large-model-based solution. Objectives are precise and realistic. Appropriate (5–7): Problem and motivation are clear, though scope or justification could be stronger. Needs improvement (3–4): Problem is somewhat vague, too broad, or weakly motivated. Hard to understand (1–2): Motivation is unclear or poorly framed. No submission (0): Motivation not provided.	/10
System Design	Outstanding (8–10): Presents a coherent and well-justified system design built around at least one open-source large language model. Components, workflow, modelling choices, and design decisions are clearly explained and technically sound. Appropriate (5–7): System design is functional and mostly well described, though some parts need clearer explanation or justification. Needs improvement (3–4): System design is incomplete, weakly justified, or only partially implemented. Hard to understand (1–2): System design is unclear or largely missing. No submission (0): System design not provided.	/10
Dataset	Outstanding (8–10): Clearly describes the dataset, knowledge sources, or other resources used, including source, relevance, structure, preprocessing, and quality considerations. The dataset choice is well aligned with the project goal. Appropriate (5–7): Adequate dataset description with minor omissions in detail or justification. Needs improvement (3–4): Basic description only; important details such as source, preprocessing, or relevance are missing. Hard to understand (1–2): Dataset description is unclear, insufficient, or weakly connected to the project. No submission (0): Dataset not described.	/10
Experiment Design	Outstanding (8–10): Experimental setup is well designed and clearly documented. At least two modelling strategies are meaningfully compared within the same task setting. The comparison is fair, justified, and suitable for answering the project questions. Appropriate (5–7): Experimental design is reasonable and includes comparison, though control, fairness, or justification could be stronger. Needs improvement (3–4): Comparison is superficial, weakly controlled, or underdeveloped. Hard to understand (1–2): Experimental design is unclear or lacks a meaningful comparison. No submission (0): Experiment design not provided.	/10
Evaluation Methodology and Results	Outstanding (8–10): Evaluation methodology is clear, appropriate, and well justified. Results are presented clearly using suitable quantitative and/or qualitative measures, tables, or figures, and directly support the project objectives. Appropriate (5–7): Evaluation is generally sound and results are reasonably clear, though detail, justification, or presentation could be improved. Needs improvement (3–4): Limited evaluation with weak methodology, unclear metrics, or insufficiently presented results. Hard to understand (1–2): Evaluation design and/or results are poorly explained, inappropriate, or difficult to interpret. No submission (0): Evaluation methodology and results not provided.	/10
Critical Analysis on Ethical Issues	Outstanding (8–10): Provides thoughtful and well-supported analysis of relevant ethical issues, such as bias, fairness, privacy, safety, misuse, or other societal implications. Limitations are clearly identified and discussed in relation to the proposed system. Appropriate (5–7): Addresses major ethical issues and limitations, though the discussion lacks depth, specificity, or critical reflection. Needs improvement (3–4): Mentions ethical concerns only briefly or in a generic way, with limited connection to the project. Hard to understand (1–2): Ethical analysis is weak, unclear, or largely irrelevant. No submission (0): Critical ethical analysis not provided.	/10

Source Code	Outstanding (8–10): Code is clean, well organised, well documented, and runnable. Core results can be reproduced with reasonable effort. Appropriate (5–7): Code is generally functional and understandable, though organisation or documentation could be improved. Needs improvement (3–4): Code is difficult to follow, insufficiently documented, or only partially reproducible. Hard to understand (1–2): Code is incomplete, poorly organised, or difficult to run. No submission (0): Code not provided.	/10
Presentation	Outstanding (8–10): The presentation is clear, concise, professional, and well structured. The group communicates the problem, system design, experimental setup, and key findings effectively within the time limit, with strong use of evidence and clear visual support. Appropriate (5–7): The presentation communicates the core ideas clearly, though explanation, structure, visual support, or timing could be improved. Needs improvement (3–4): The presentation is understandable but lacks focus, clarity, or effective organisation. Hard to understand (1–2): The presentation is unclear, poorly structured, or fails to communicate the core project ideas. No submission (0): No presentation delivered.	/10
Q&A	Outstanding (5): Answers questions accurately, confidently, and with clear understanding of the project design, methods, results, and limitations. Appropriate (3–4): Answers most questions adequately, though some responses may lack precision, depth, or confidence. Needs improvement (2): Responses show limited understanding of parts of the project or fail to address questions clearly. Hard to understand (1): Responses are vague, inaccurate, or show little understanding of the project. No submission (0): No meaningful participation in the Q&A.	/5
Live Demonstration	Outstanding (13–15): The system runs successfully in class and demonstrates realistic, well-prepared use cases. The demo clearly shows the main functionality, highlights differences between modelling strategies where appropriate, and includes meaningful observations such as strengths, limitations, or failure cases. Appropriate (10–12): The demo is functional and shows the main system capabilities, though preparation, clarity, or depth could be improved. Needs improvement (6–9): The demo is only partially successful, limited in scope, or does not clearly show the intended contribution. Hard to understand (1–5): The demo is poorly prepared, unsuccessful, or fails to show meaningful system behaviour. No submission (0): No live demonstration provided.	/15